

Practical no 11

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Output:-

```
print("*" * 40)

print("    Bus Reservation System")

print("*" * 40)

total_rows = int(input("Enter number of rows in the bus: "))
total_columns = int(input("Enter number of columns in the bus: "))

bus_layout = []

for r in range(total_rows):

    rows = []

    for c in range(total_columns):

        rows.append("O")

    bus_layout.append(rows)

print(f"\nBus layout created: {total_rows} rows * {total_columns} columns.")

print("All seats are currently Open (O).\n")

while True:

    print("*" * 40)

    print("1. Display Bus Layout.")

    print("2. Reserve a Seat.")

    print("3. Cancel a Reservation.")

    print("4. Count Available / Reserved Seats.")

    print("5. Check a Specific Seat Status.")

    print("6. Exit.")

    print("'" * 40)
```

```

choice = input("Enter your choice (1-6): ").strip

if choice == "1":

    print("\nCurrent Bus Layout:")

    print("(Rows top to bottom = Rows 1 to", total_rows, ")\n")


    for r in range(len(bus_layout)):

        print(f"Row {r + 1}: ", end="")

        for c in range(len(bus_layout[r])):

            print(bus_layout[r][c], end=" ")

        print()

    print()

elif choice == "2":

    row_num = int( input(f"Enter row number (1 to {total_rows}): ") ) - 1

    col_num = int( input(f"Enter seat number (1 to {total_columns}): ") ) - 1

    if 0 <= row_num < total_rows and 0 <= col_num < total_columns:

        if bus_layout[row_num][col_num] == "X":

            print("This seat is already reserved.\n")

        else:

            bus_layout[row_num][col_num] = "X"

            print(

                f"Seat at Row {row_num + 1}, "

                f"Seat {col_num + 1} reserved successfully.\n"

            )

    else:

        print("Invalid row or seat number.\n")

```

```

elif choice == "3":

    row_num = int(input(f"Enter row number (1 to {total_rows}): ")) - 1
col_num = int( input(f"Enter seat number (1 to {total_columns}): ")) - 1
if 0 <= row_num < total_rows and 0 <= col_num < total_columns:

    if bus_layout[row_num][col_num] == "O":

        print("This seat is already open.\n")

    else:

        bus_layout[row_num][col_num] = "O"

        print(

            f"Reservation cancelled for Row {row_num + 1}, "

            f"Seat {col_num + 1}.\n"

        )

    else:

        print("Invalid row or seat number.\n")

elif choice == "4":

    open_count = 0

    reserved_count = 0

for r in range(len(bus_layout)):

    for c in range(len(bus_layout[r])):

        if bus_layout[r][c] == "O":

            open_count += 1

        else:

            reserved_count += 1

```

```
print(f"\nTotal Seats: {total_rows * total_columns}")

print(f"Open Seats: {open_count}")

print(f"Reserved Seats: {reserved_count}\n")

elif choice == "5":

    row_num = int(input(f"Enter row number (1 to {total_rows}): ")) - 1
    col_num = int(input(f"Enter seat number (1 to {total_columns}): ")) - 1

    if 0 <= row_num < total_rows and 0 <= col_num < total_columns:

        status = bus_layout[row_num][col_num]

        if status == "O":

            print(

                f"Seat Row {row_num + 1}, Seat {col_num + 1} "

                f"is Open.\n"

            )

        else:

            print(

                f"Seat Row {row_num + 1}, Seat {col_num + 1} "

                f"is Reserved.\n"

            )

        else:

            print("Invalid row or seat number.\n")

elif choice == "6":

    print("Exiting program. Thank you.")

    break

else:
```

```
print("Invalid choice. Please enter a number between 1 and 6.\n")
```

Output:-

Problems Output Debug Console **Terminal** Ports

```
PS C:\Users\Admin\revati> & 'C:\Users\Admin\AppData\Roaming\uv\python\cpython-3.14-5100' '--' 'c:\Users\Admin\revati\practical 11.py'
```

```
*****
```

Bus Reservation System

```
*****
```

Enter number of rows in the bus: 10

Enter number of columns in the bus: 2

Bus layout created: 10 rows * 2 columns.

All seats are currently Open (0).

```
*****
```

1. Display Bus Layout.
2. Reserve a Seat.
3. Cancel a Reservation.
4. Count Available / Reserved Seats.
5. Check a Specific Seat Status.
6. Exit.

```
*****
```

Enter your choice (1-6): 1

Current Bus Layout:

(Rows top to bottom = Rows 1 to 10)

Row 1: 0 0

Row 2: 0 0

Row 3: 0 0

Row 4: 0 0

Row 5: 0 0

Row 6: 0 0

Row 7: 0 0

Row 8: 0 0

Row 9: 0 0

Row 10: 0 0

```
*****
```

1. Display Bus Layout.
2. Reserve a Seat.
3. Cancel a Reservation.
4. Count Available / Reserved Seats.
5. Check a Specific Seat Status.
6. Exit.

```
*****
```

Enter your choice (1-6): 2

Enter row number (1 to 10): 5

Enter seat number (1 to 2): 1

Seat at Row 5, Seat 1 reserved successfully.

```
*****
```

1. Display Bus Layout.
2. Reserve a Seat.
3. Cancel a Reservation.
4. Count Available / Reserved Seats.
5. Check a Specific Seat Status.

...

Problems Output Debug Console **Terminal** Ports

```
5. Check a Specific Seat Status.
6. Exit.
*****
Enter your choice (1-6): 2
Enter row number (1 to 10): 5
Enter seat number (1 to 2): 1
Seat at Row 5, Seat 1 reserved successfully.
```

```
*****
1. Display Bus Layout.
2. Reserve a Seat.
3. Cancel a Reservation.
4. Count Available / Reserved Seats.
5. Check a Specific Seat Status.
6. Exit.
*****
Enter your choice (1-6): 2
Enter row number (1 to 10): 7
Enter seat number (1 to 2): 2
Seat at Row 7, Seat 2 reserved successfully.
```

```
*****
1. Display Bus Layout.
2. Reserve a Seat.
3. Cancel a Reservation.
4. Count Available / Reserved Seats.
5. Check a Specific Seat Status.
6. Exit.
*****
Enter your choice (1-6): 4
```

```
Total Seats: 20
Open Seats: 18
Reserved Seats: 2
```

```
*****
1. Display Bus Layout.
2. Reserve a Seat.
3. Cancel a Reservation.
4. Count Available / Reserved Seats.
5. Check a Specific Seat Status.
6. Exit.
*****
Enter your choice (1-6): 4
```

```
Total Seats: 20
Open Seats: 18
Reserved Seats: 2
```

```
*****
1. Display Bus Layout.
2. Reserve a Seat.
3. Cancel a Reservation.
4. Count Available / Reserved Seats.
```

Problems Output Debug Console **Terminal** Ports

5. Check a Specific Seat Status.

6. Exit.

Enter your choice (1-6): 4

Total Seats: 20

Open Seats: 18

Reserved Seats: 2

1. Display Bus Layout.

2. Reserve a Seat.

3. Cancel a Reservation.

4. Count Available / Reserved Seats.

5. Check a Specific Seat Status.

6. Exit.

Enter your choice (1-6): 5

Enter row number (1 to 10): 7

Enter seat number (1 to 2): 2

Seat Row 7, Seat 2 is Reserved.

1. Display Bus Layout.

2. Reserve a Seat.

3. Cancel a Reservation.

4. Count Available / Reserved Seats.

5. Check a Specific Seat Status.

6. Exit.

Enter your choice (1-6): 6

Exiting program. Thank you.

PS C:\Users\Admin\revati> ^C